



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECR 1213

Network Communication

Task 4 – Making the Connections - LAN and WAN

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Table Of Content

1.0 Identify the work areas on floor plan	3
1.1 Ground Floor	3
1.2 First Floor	5
2.0 The distribution and connection of devices	7
2.1 Network Distribution Diagram	7
2.2 Room Level Network Distribution Diagram	12
2.2.1 Ground Floor	12
2.2.1.1 Cisco Networking Lab	14
2.2.1.2 General Purpose Lab	16
2.2.1.3 Lecturer's Office	18
2.2.2 First Floor	19
2.2.2.1 General Purpose Lab	21
2.2.2.2 Embedded Lab	23
2.2.2.3 Video Conferencing Room	25
2.2.2.4 Hybrid Room	26
2.2.2.5 Printing Area & Small Meeting Room	27
3.0 Number of connections, patch cords and switch ports needed	28
3.1 Ground Floor	28
3.2 First Floor	29
3.3 Summary of both floor	30
4.0 Identify cables types and length	32
4.1 CAT6 twisted cable	32
4.2 Fiber Optic Cables	33
4.3 Summary of Cables Used	34
5.0 Meeting Minutes	35
5.1 Meeting Minutes#1	35
5.2 Meeting Minutes#2	37

1.0 Identify the work areas on floor plan

1.1 Ground Floor

The ground floor of the Faculty of Computing building housed diverse rooms from leisure to hand-on labs and even a server room. Based on the floor plan we designed in Task 1, we have identified four primary work areas that require distinct network connectivity strategies.

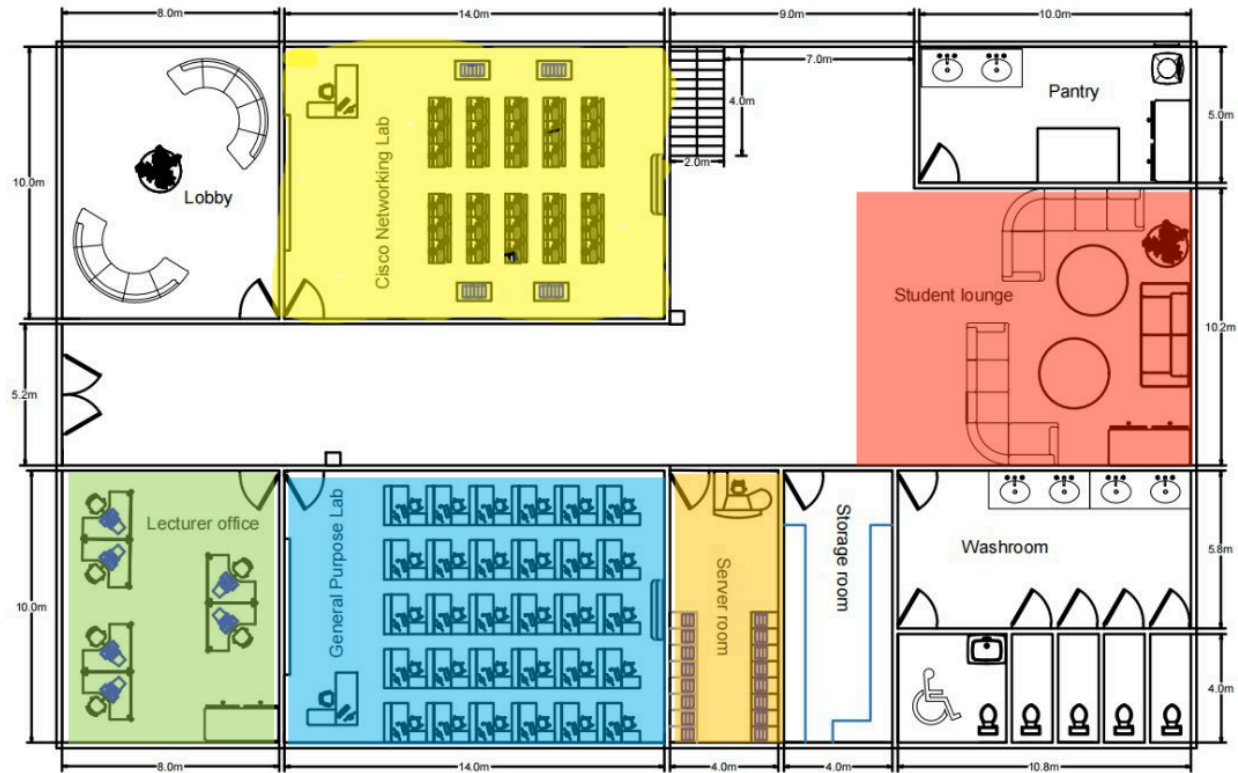


Figure 1.0: Work Area at Ground Floor

Work Area 1, as shown in Figure 1.0, is the **Cisco Networking Lab** located on the upper side of the floor plan of the ground floor. This lab is designed to support hands-on networking practice and technical instruction. It is equipped with 30 student work stations and 4 sets of switches and routers in the room span of 14 meters in length and 10 meters in width. Therefore, this area requires high-density cabling to support every student in the room and has reliable access to the faculty network during their hands-on lab practical class and exams.

Work Area 2, situated on the opposite side of Cisco Networking Lab, is the **General Purpose Lab**. It serves as a flexible computer or project room that can accommodate classes, assignments, and general computing tasks. It is equipped with 30 student workstations arranged in rows and one teaching desk. This area will have to require high-density cabling too to support 31 wired connections, ensuring every student has reliable access to the faculty network during exams and practical classes.

Work Area 3, situated in the bottom-left corner of the floor plan, is the **Lecturer's Office**. This room provides a private and comfortable professional environment for academic work and consultations. It is furnished with desks, chairs, cabinets, and decorative plants. The office is equipped with six personal cubicles, each featuring a full workstation (monitor, CPU, and keyboard), allowing lecturers to prepare study materials and conduct administrative work efficiently.

Work Area 4, located centrally along the bottom wall, is the **Server Room** (Telecommunications Room). It houses essential technical infrastructure, including servers, network switches, routers, and a Wi-Fi access point. Designed with controlled access to ensure the protection and maintenance of sensitive equipment, the room contains two rows of eight servers (16 total). This configuration is designed to handle high-density networking for the 42m x 26m two-story building, supporting the labs, meeting room, hybrid room, and video conference room.

Work Area 5, located on the right side of the first floor, is the **Student Lounge**. This dedicated rest and social zone is equipped with L-shaped and U-shaped sofas, round tables, and soft seating, offering students a comfortable place to relax, study informally, or interact with peers. Although there are no electronic devices stationed in this area, reliable network access is essential, as students frequently use the space for collaborative group projects on their personal devices.

1.2 First Floor

The First Floor of the Faculty of Computing building is designed to support specialized technical education and collaborative learning. Based on the floor plan designed in Task 1, we have identified six primary work areas that require distinct network connectivity strategies.

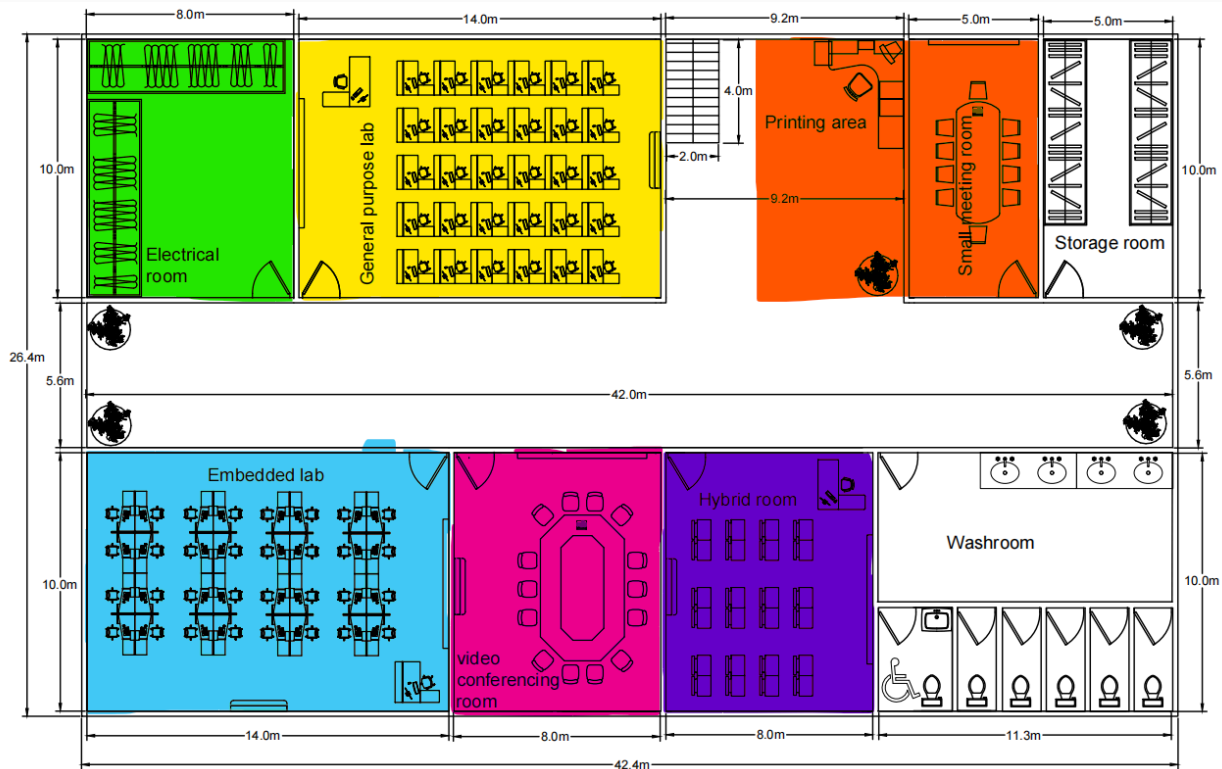


Figure 1.2: Work Area at First Floor

Work Area 6, as shown in Figure 1.2, is the **General Purpose Lab** located on the upper side of the floor plan. This lab is a spacious classroom designed for theory-based lessons and general computing tasks. It is equipped with 30 student workstations arranged in rows and one teaching desk. Consequently, this area requires high-density cabling to support 31 wired connections, ensuring every student has reliable access to the faculty network during exams and practical classes.

Work Area 7, situated on the bottom left of the floor plan, is the **Embedded Lab**. This facility is designed for hands-on technical work, featuring multiple lab desks arranged for group or individual practical activities. It houses 30 workstations for students. Unlike standard labs, the layout here is clustered for collaboration, requiring cable management that runs to the center of workstation islands rather than just along the walls.

Work Area 8, located centrally on the bottom side, is the **Video Conferencing Room**. This room features a central conference table and is equipped to support remote meetings. While it has fewer devices than the labs, it requires a high-priority, high-bandwidth connection to ensure

smooth, lag-free video streaming. A dedicated Wireless Access Point (WAP) is also essential here to support staff laptops.

Work Area 9 is the **Hybrid Room**, located on the bottom right of the building. This flexible teaching environment features neatly aligned desks and a smartboard. It contains 30 workstations. Although this room is located approximately 40 meters from the Electrical Room, we will run direct horizontal cabling through the corridor ceiling trays to connect these workstations to the main distribution rack, avoiding the need for additional active equipment.

Work Area 10 encompasses the support facilities located on the upper right side:

- **Printing Area:** Centrally located to be accessible to students and staff. It requires a wired data port for the shared network printer.
- **Small Meeting Room:** Designed for discussions and group projects. This room requires robust wireless coverage to support students and staff working on laptops.

Work Area 11 is the **Electrical Room**, situated on the top left corner. We have designated this secure space to serve as the Intermediate Distribution Facility (IDF) for the entire First Floor. It will house the centralized network rack containing the switches that serve all labs, including the distant Hybrid Room and Printing Area.

2.0 The distribution and connection of devices

2.1 Network Distribution Diagram

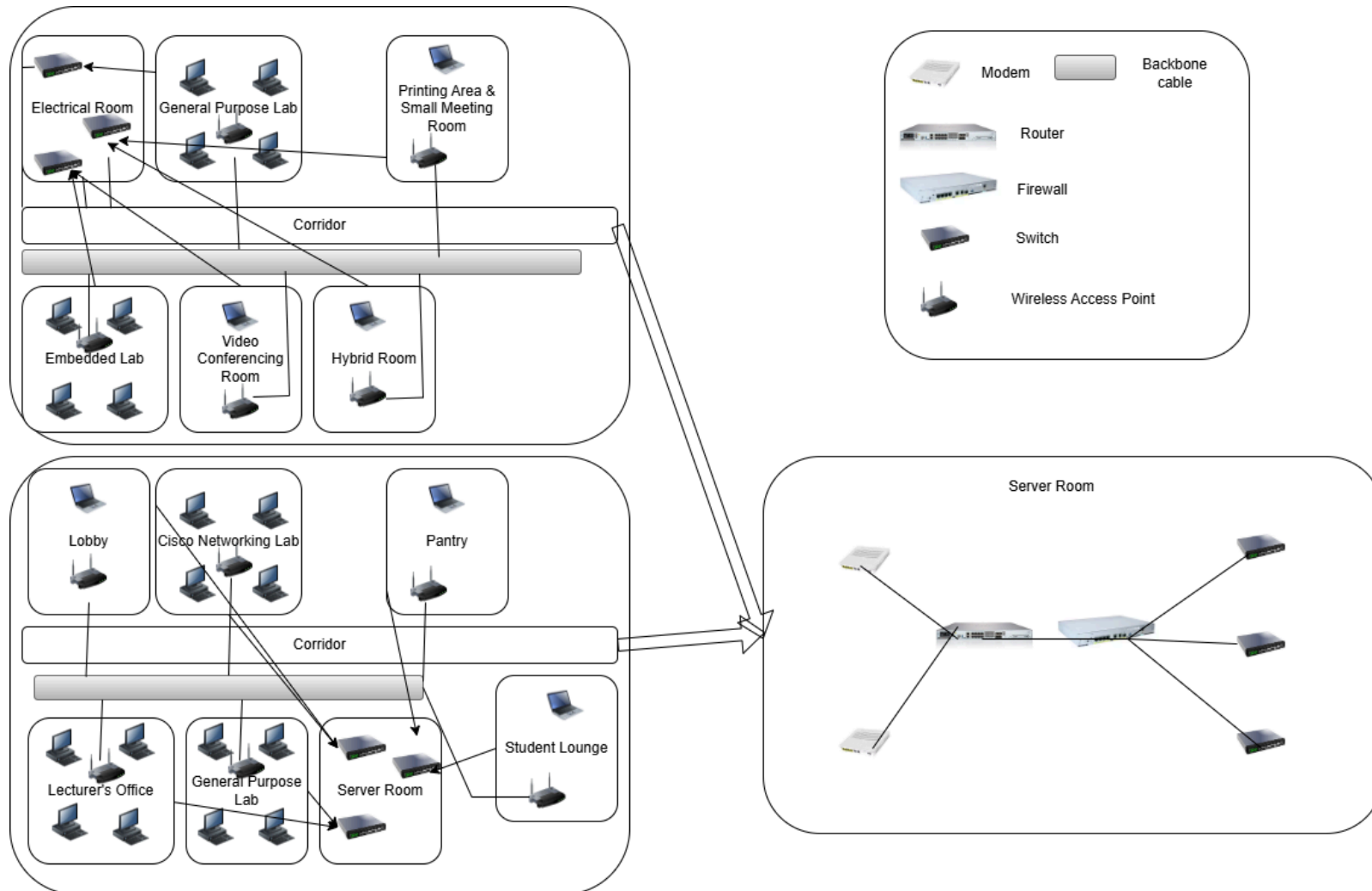


Figure 2.1.1: Network Distribution Diagram

This section explains the overall network structure, known as the Logical Topology, detailing the connections between the floors, the work areas, and the central distribution points.

The design utilizes a two-tier Hierarchical Star Topology to ensure scalability, ease of management, and high performance across the building.

1. Core Distribution Facilities

The network relies on two centralized distribution points:

- Main Distribution Facility (MDF): Located in the Server Room on the Ground Floor. This room houses the main router, firewall, servers, and the central switching equipment that provides connectivity to the external Internet Service Provider (ISP) and the campus network.
- Intermediate Distribution Facility (IDF): Located in the Electrical Room on the First Floor. This IDF centralizes the network switches and patch panels necessary to serve all work areas on the First Floor, reducing cable runs and simplifying maintenance.

2. Backbone Connectivity (Inter-Floor)

The critical link between the two floors is the network backbone:

- Connection: The MDF (Server Room) and the IDF (Electrical Room) are linked using a dedicated Fiber Optic Cable.
- Role: This backbone connection is essential for handling the aggregated traffic from all labs and offices on the First Floor, providing the necessary high bandwidth and reliability to the main router/server infrastructure on the Ground Floor.

3. Horizontal Connectivity (Intra-Floor)

Within each floor, connectivity follows a star pattern:

- Ground Floor: All work areas (Cisco Networking Lab, General Purpose Lab, Lecturer Office) and supporting areas connect directly to the central switch and patch panel located within the Server Room (MDF).
- First Floor: All work areas (General Purpose Lab, Embedded Lab, Hybrid Room, etc.) and their respective workstations and Wireless Access Points (WAPs) run dedicated CAT6 cabling to the centralized patch panels and switches located within the Electrical Room (IDF).
- Topology Detail: The physical cabling for all work areas runs horizontally above the ceiling through dedicated corridor trunking to reach the IDF/MDF, as shown in the floor diagrams.

This hierarchy ensures that high-density areas like the Labs have reliable, dedicated pathways back to the core network infrastructure.

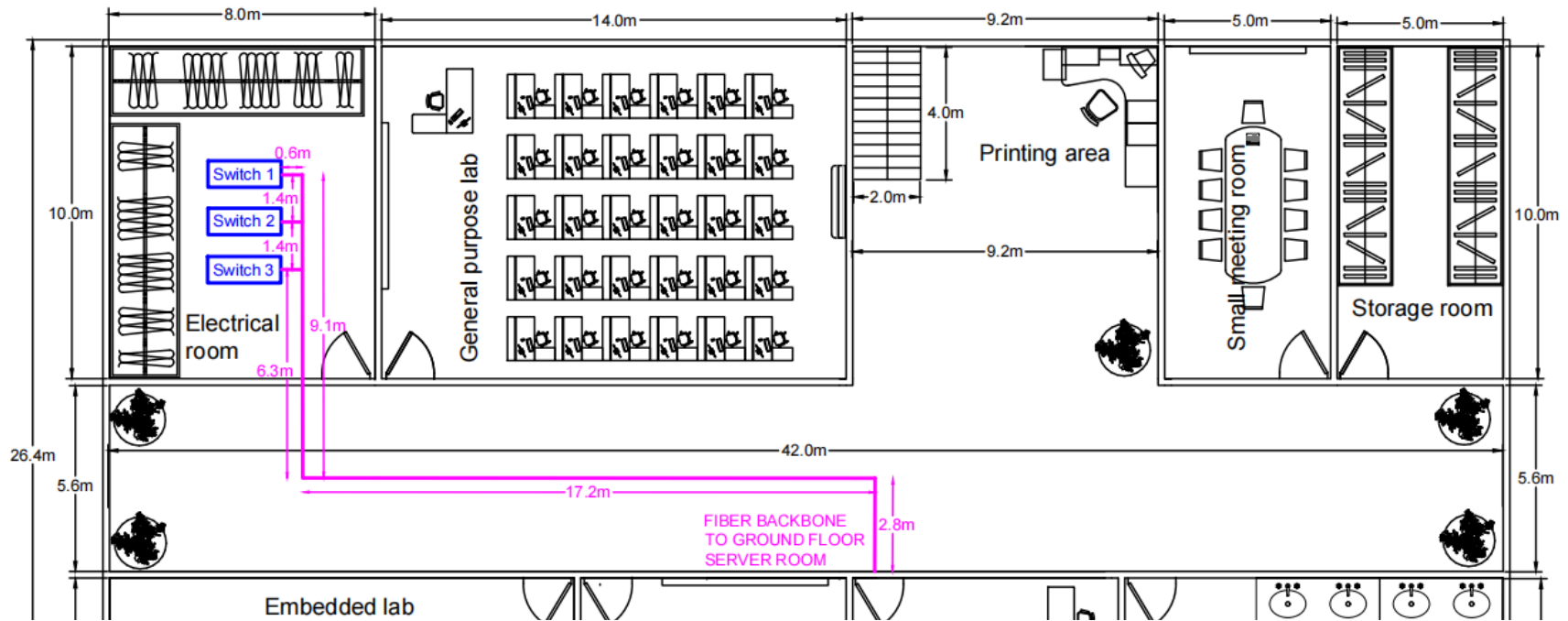


Figure 2.1.2: First Floor Fiber Backbone

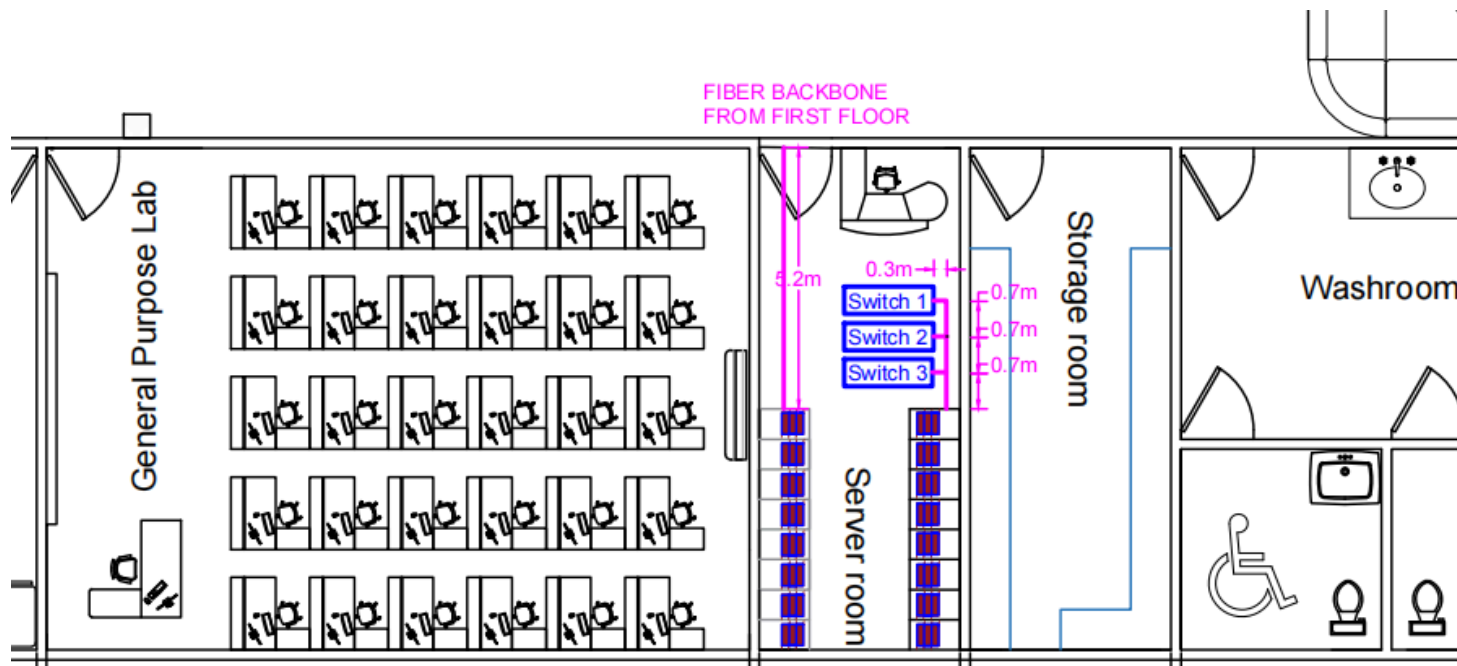


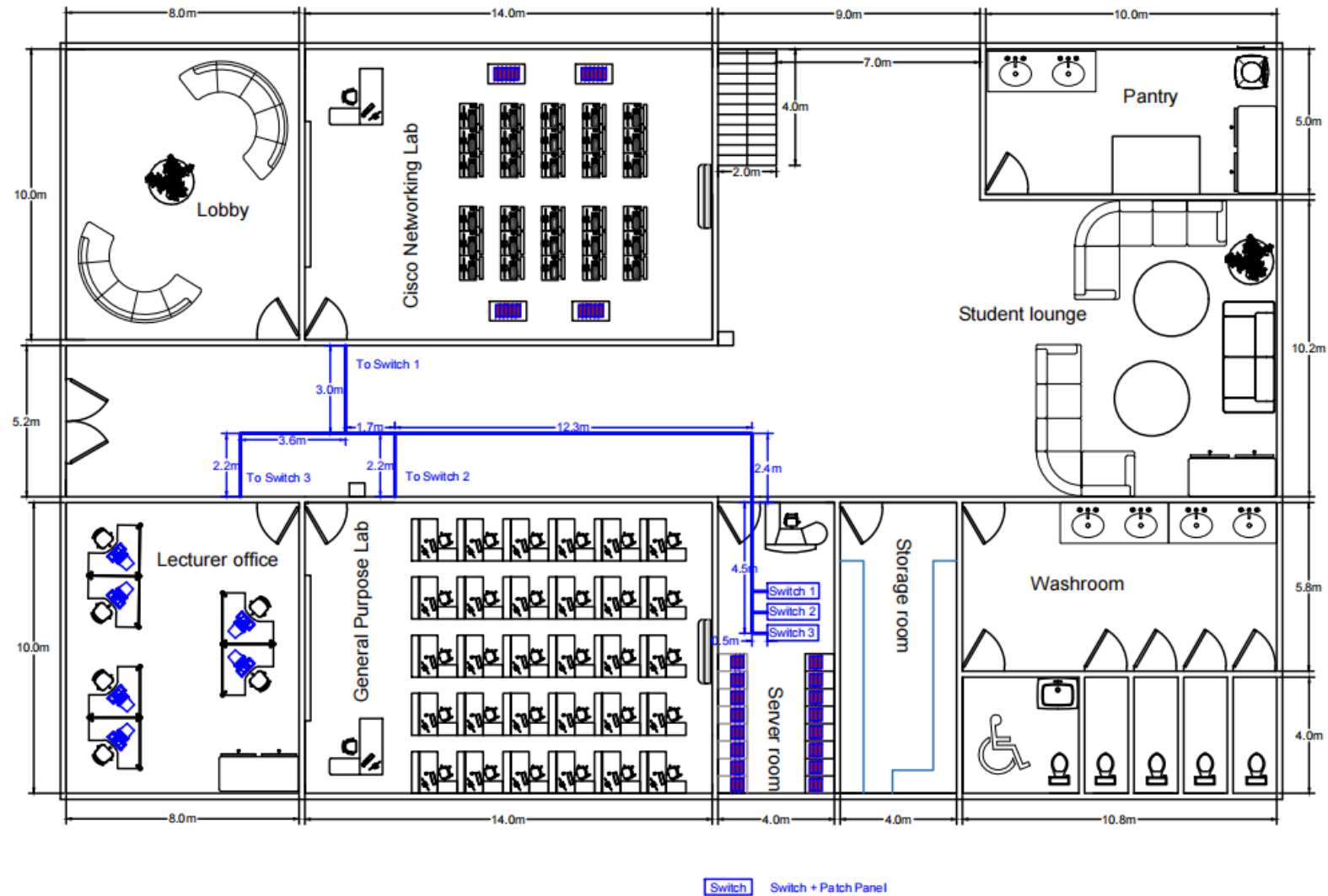
Figure 2.1.3 : Ground Floor Fiber Backbone

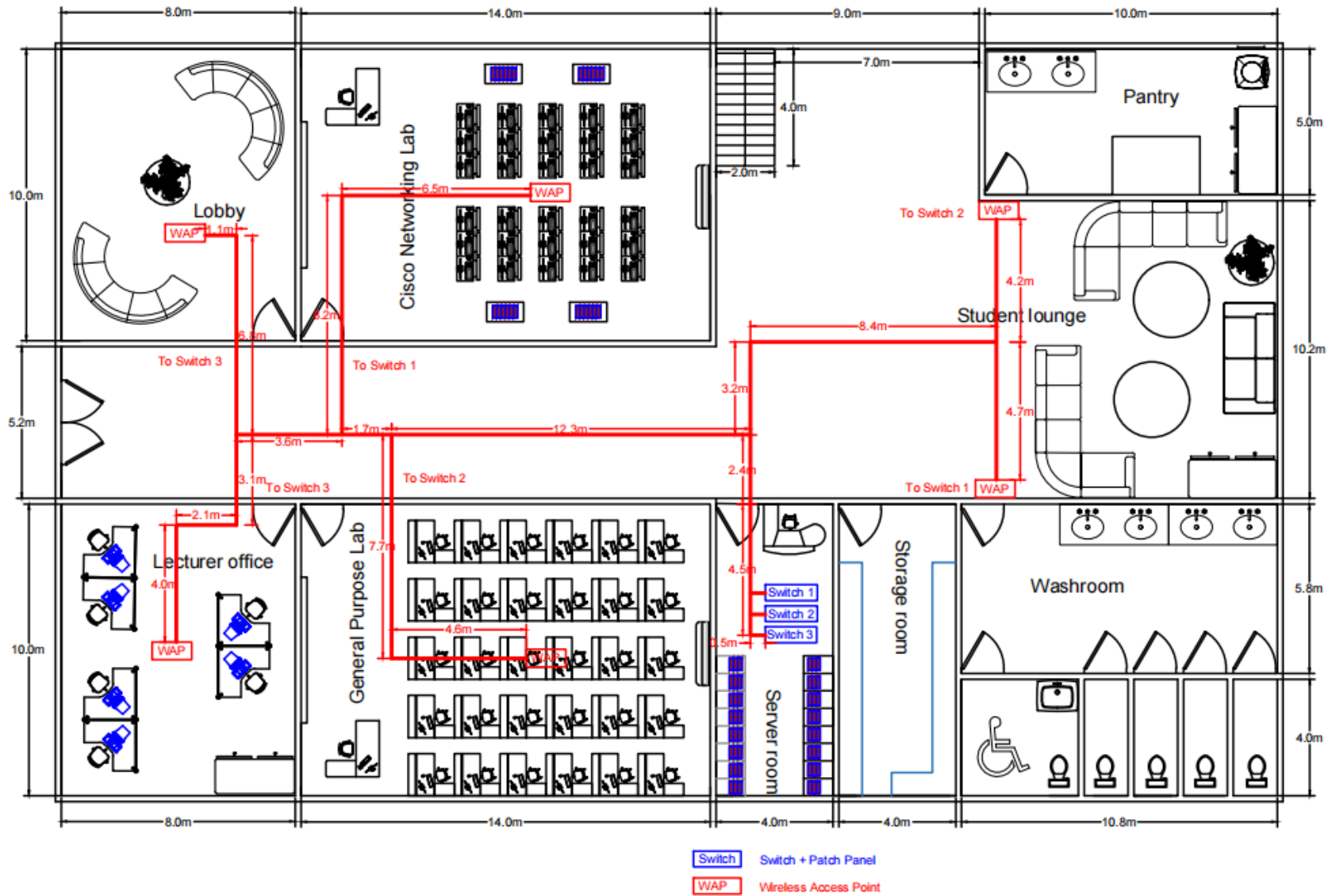
Connection to Server	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Ground Floor Switch 1	Fiber Optic	$0.3 + 0.7 + 0.7 + 0.7 = 2.4$	0.5	2.9
Ground Floor Switch 2	Fiber Optic	$0.3 + 0.7 + 0.7 = 1.7$	0.5	2.2

Ground Floor Switch 3	Fiber Optic	$0.3 + 0.7 = 1.0$	0.5	1.5
First Floor Switch 1	Fiber Optic	$0.6 + 9.1 + 17.2 + 2.8 + 5.2 = 34.9$	1.7	36.6
First Floor Switch 2	Fiber Optic	$0.6 + 1.4 + 6.3 + 17.2 + 2.8 + 5.2 = 33.5$	1.7	35.2
First Floor Switch 3	Fiber Optic	$0.6 + 6.3 + 17.2 + 2.8 + 5.2 = 32.1$	1.7	33.8
Total		105.6	6.6	112.2

2.2 Room Level Network Distribution Diagram

2.2.1 Ground Floor





2.2.1.1 Cisco Networking Lab

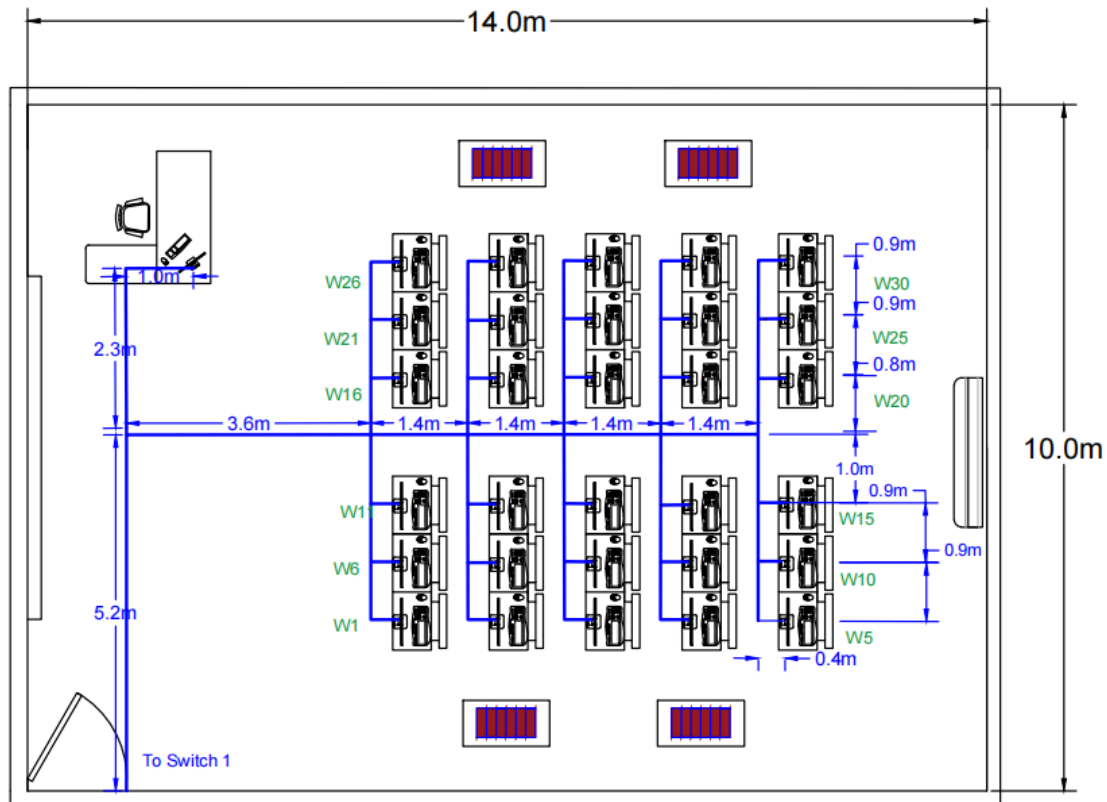


Figure 2.2.1.1: Room Level Cabling - Cisco Network Cabling

Connection to Switch 1 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Workstation W1	Cat 6	$0.4 + 0.9 + 0.9 + 1.0 + 3.6 + 5.2 = 12.0$	0.75	12.75
Workstation W2	Cat 6	$12 + 1.4 = 13.4$	0.75	14.15
Workstation W3	Cat 6	$13.4 + 1.4 = 14.8$	0.75	15.55
Workstation W4	Cat 6	$14.8 + 1.4 = 16.2$	0.75	16.95
Workstation W5	Cat 6	$16.2 + 1.4 = 17.6$	0.75	18.35
Workstation W6	Cat 6	$0.4 + 0.9 + 1.0 + 3.6 + 5.2 = 11.1$	0.75	11.85
Workstation W7	Cat 6	$11.1 + 1.4 = 12.5$	0.75	13.25
Workstation W8	Cat 6	$12.5 + 1.4 = 13.9$	0.75	14.65

Workstation W9	Cat 6	$13.9 + 1.4 = 15.3$	0.75	16.05
Workstation W10	Cat 6	$15.3 + 1.4 = 16.7$	0.75	17.45
Workstation W11	Cat 6	$0.4 + 1.0 + 3.6 + 5.2 = 10.2$	0.75	10.95
Workstation W12	Cat 6	$10.2 + 1.4 = 11.6$	0.75	12.35
Workstation W13	Cat 6	$11.6 + 1.4 = 13$	0.75	13.75
Workstation W14	Cat 6	$13 + 1.4 = 14.4$	0.75	15.15
Workstation W15	Cat 6	$14.4 + 1.4 = 15.8$	0.75	16.55
Workstation W16	Cat 6	$0.4 + 0.8 + 3.6 + 5.2 = 10$	0.75	10.75
Workstation W17	Cat 6	$10 + 1.4 = 11.4$	0.75	12.15
Workstation W18	Cat 6	$11.4 + 1.4 = 12.8$	0.75	13.55
Workstation W19	Cat 6	$12.8 + 1.4 = 14.2$	0.75	14.95
Workstation W20	Cat 6	$14.2 + 1.4 = 15.6$	0.75	16.35
Workstation W21	Cat 6	$0.4 + 0.9 + 0.8 + 3.6 + 5.2 = 10.9$	0.75	11.65
Workstation W22	Cat 6	$10.9 + 1.4 = 12.3$	0.75	13.05
Workstation W23	Cat 6	$12.3 + 1.4 = 13.7$	0.75	14.45
Workstation W24	Cat 6	$13.7 + 1.4 = 15.1$	0.75	15.85
Workstation W25	Cat 6	$15.1 + 1.4 = 16.5$	0.75	17.25
Workstation W26	Cat 6	$0.4 + 0.9 + 0.9 + 0.8 + 3.6 + 5.2 = 11.8$	0.75	12.55
Workstation W27	Cat 6	$11.8 + 1.4 = 12.8$	0.75	13.55
Workstation W28	Cat 6	$12.8 + 1.4 = 14.2$	0.75	14.95
Workstation W29	Cat 6	$14.2 + 1.4 = 15.6$	0.75	16.35
Workstation W30	Cat 6	$15.6 + 1.4 = 17$	0.75	17.75
Teaching Desk	Cat 6	$1.0 + 2.3 + 5.2 = 8.5$	0.75	9.25
WAP	Cat 6	$6.5 + 6.2 + 1.7 + 12.3 + 2.4 + 4.5 + 0.5 = 34.1$	2.7	36.8
Total		455	25.2	480.2

2.2.1.2 General Purpose Lab

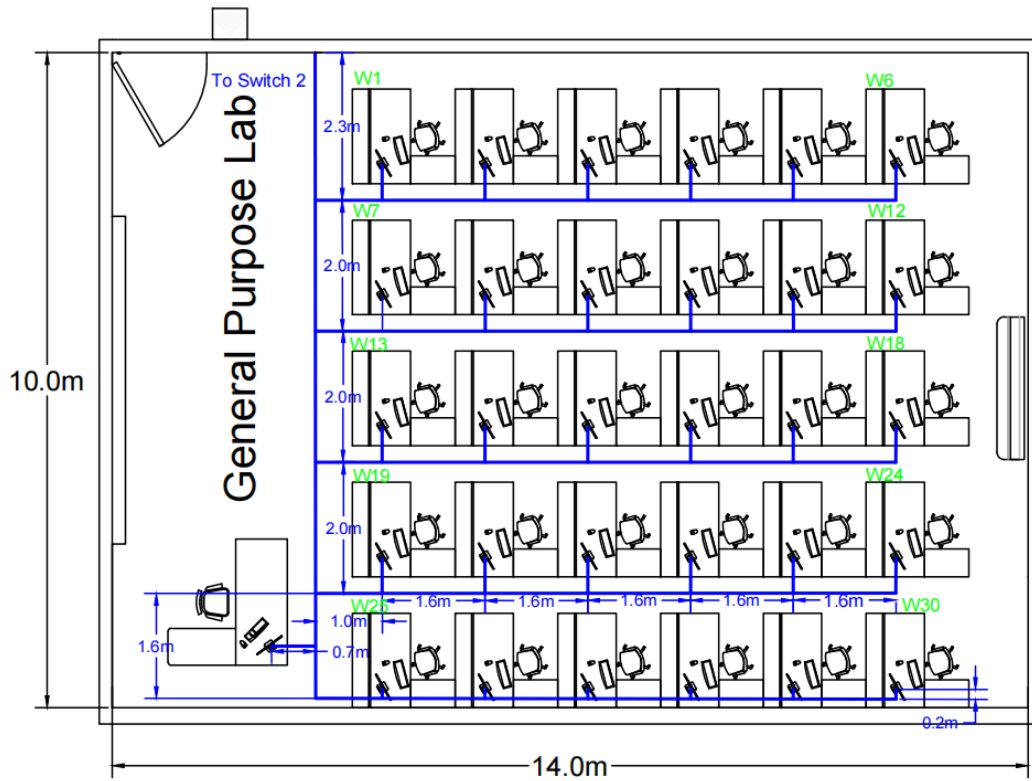


Figure 2.2.1.2: Room Level Cabling - General Purpose Lab

Connection to Switch 1 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Workstation W1	Cat 6	$0.2 + 1.0 + 2.3 = 3.5$	0.75	4.25
Workstation W2	Cat 6	$3.5 + 1.6 = 5.1$	0.75	5.85
Workstation W3	Cat 6	$5.1 + 1.6 = 6.7$	0.75	7.45
Workstation W4	Cat 6	$6.7 + 1.6 = 8.3$	0.75	9.05
Workstation W5	Cat 6	$8.3 + 1.6 = 9.9$	0.75	10.65
Workstation W6	Cat 6	$9.9 + 1.6 = 11.5$	0.75	12.25
Workstation W7	Cat 6	$3.5 + 2.0 = 5.5$	0.75	6.25
Workstation W8	Cat 6	$5.1 + 2.0 = 7.1$	0.75	7.85

Workstation W9	Cat 6	$7.1 + 2.0 = 9.1$	0.75	9.85
Workstation W10	Cat 6	$8.3 + 2.0 = 10.3$	0.75	11.05
Workstation W11	Cat 6	$10.3 + 2.0 = 12.3$	0.75	13.05
Workstation W12	Cat 6	$11.5 + 2.0 = 13.5$	0.75	14.25
Workstation W13	Cat 6	$5.5 + 2.0 = 7.5$	0.75	8.25
Workstation W14	Cat 6	$7.1 + 2.0 = 9.1$	0.75	9.85
Workstation W15	Cat 6	$9.1 + 2.0 = 11.1$	0.75	11.85
Workstation W16	Cat 6	$10.3 + 2.0 = 12.3$	0.75	13.05
Workstation W17	Cat 6	$12.3 + 2.0 = 14.3$	0.75	15.05
Workstation W18	Cat 6	$13.5 + 2.0 = 15.5$	0.75	16.25
Workstation W19	Cat 6	$7.5 + 2.0 = 9.5$	0.75	10.25
Workstation W20	Cat 6	$9.1 + 2.0 = 11.1$	0.75	11.85
Workstation W21	Cat 6	$11.1 + 2.0 = 13.1$	0.75	13.85
Workstation W22	Cat 6	$12.3 + 2.0 = 14.3$	0.75	15.05
Workstation W23	Cat 6	$14.3 + 2.0 = 16.3$	0.75	17.05
Workstation W24	Cat 6	$15.5 + 2.0 = 17.5$	0.75	18.25
Workstation W25	Cat 6	$9.5 + 1.6 = 11.1$	0.75	11.85
Workstation W26	Cat 6	$11.1 + 1.6 = 12.7$	0.75	13.45
Workstation W27	Cat 6	$12.7 + 1.6 = 14.3$	0.75	15.05
Workstation W28	Cat 6	$14.3 + 1.6 = 15.9$	0.75	16.65
Workstation W29	Cat 6	$16.3 + 1.6 = 17.9$	0.75	18.65
Workstation W30	Cat 6	$17.5 + 1.6 = 19.1$	0.75	19.85
Teaching Desk	Cat 6	$0.7 + 0.6 + 6 + 2.3 = 9.6$	0.75	10.35
WAP	Cat 6	$4.6 + 7.7 + 12.3 + 2.4 + 4.5 + 0.5 = 32.1$	2.7	34.8
Total		387.1	25.95	413.05

2.2.1.3 Lecturer's Office

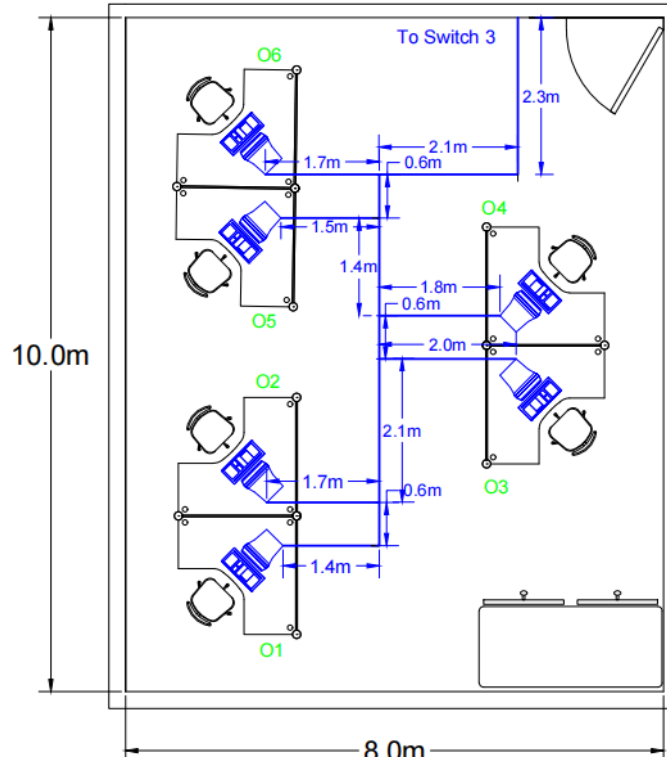
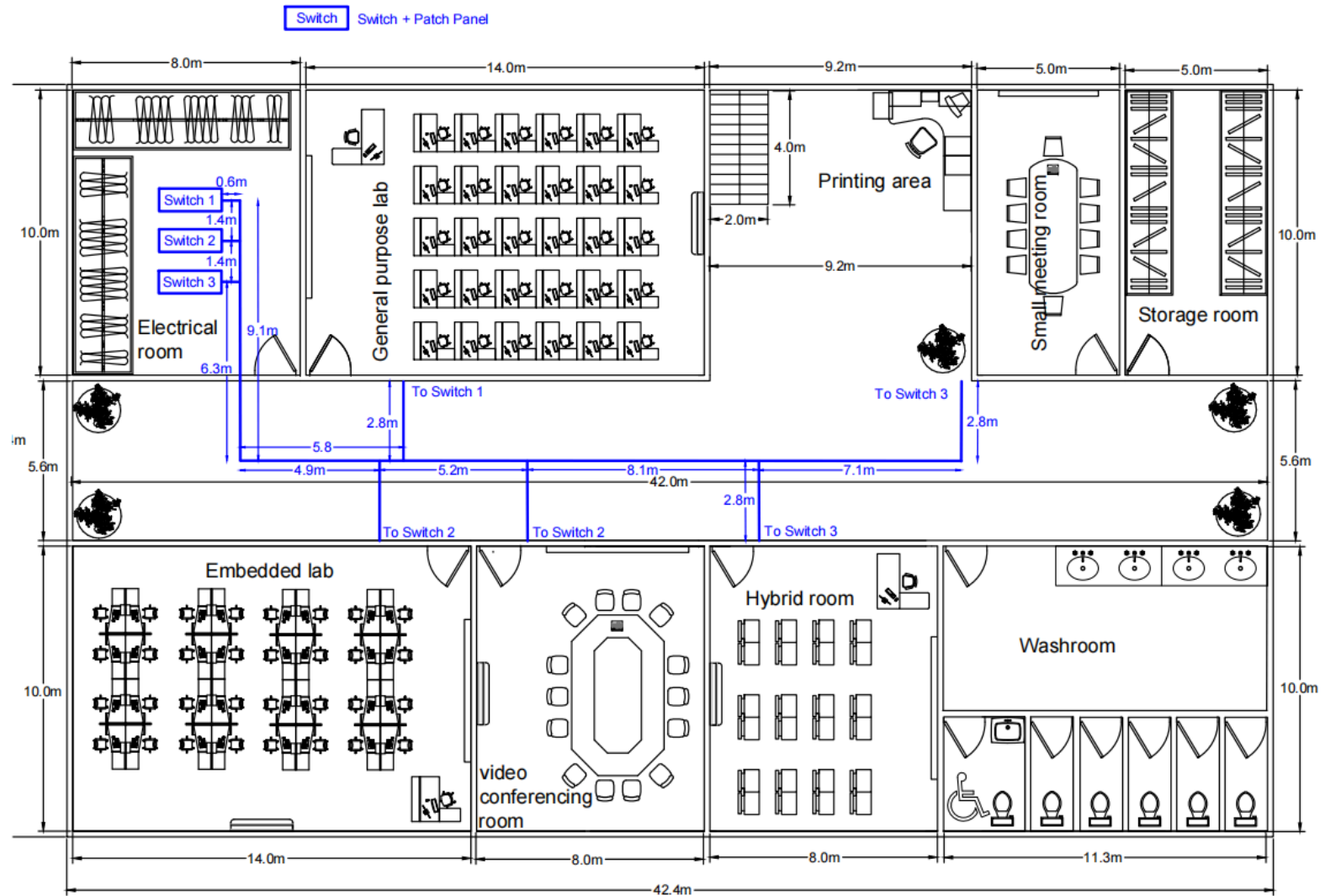
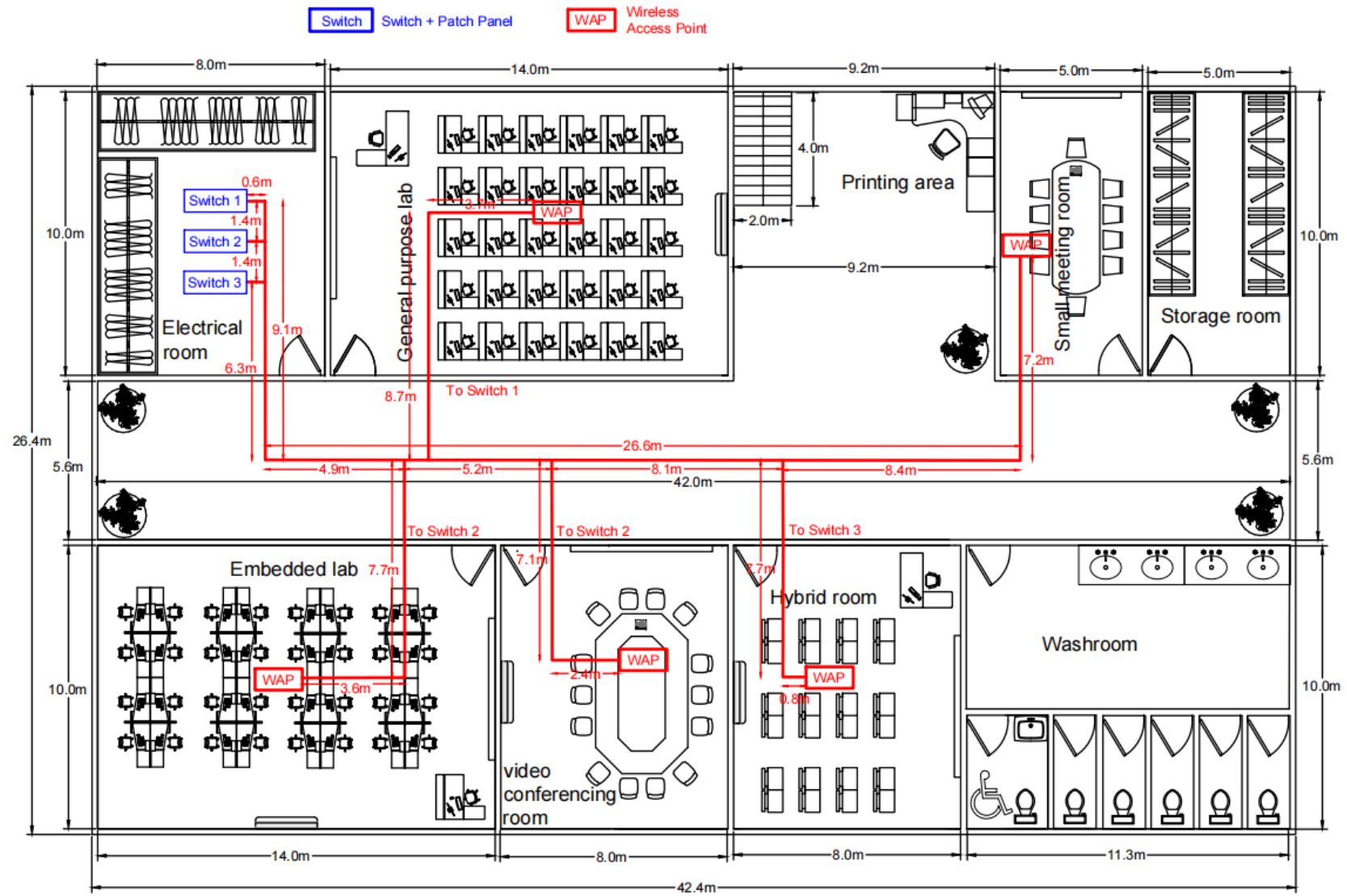


Figure 2.2.1.3: Room Level Cabling - Lecturer's Office

Connection to Switch 1 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Office Desk 1	Cat 6	$1.4 + 0.6 + 2.1 + 0.6 + 1.4 + 0.6 + 2.1 + 2.3 = 11.1$	0.75	11.85
Office Desk 2	Cat 6	$1.7 + 2.1 + 0.6 + 1.4 + 0.6 + 2.1 + 2.3 = 10.8$	0.75	11.55
Office Desk 3	Cat 6	$2 + 0.6 + 1.4 + 0.6 + 2.1 + 2.3 = 9.0$	0.75	9.75
Office Desk 4	Cat 6	$1.8 + 1.4 + 0.6 + 2.1 + 2.3 = 8.2$	0.75	8.95
Office Desk 5	Cat 6	$1.5 + 0.6 + 2.1 + 2.3 = 6.5$	0.75	7.25
Office Desk 6	Cat 6	$1.7 + 2.1 + 2.3 = 6.1$	0.75	6.85
WAP	Cat 6	$4.0 + 2.1 + 3.1 + 3.6 + 1.7 + 12.3 + 2.4 + 4.5 + 1.5 = 35.2$	2.7	37.9
Total		86.9	7.2	94.1

2.2.2 First Floor





2.2.2.1 General Purpose Lab

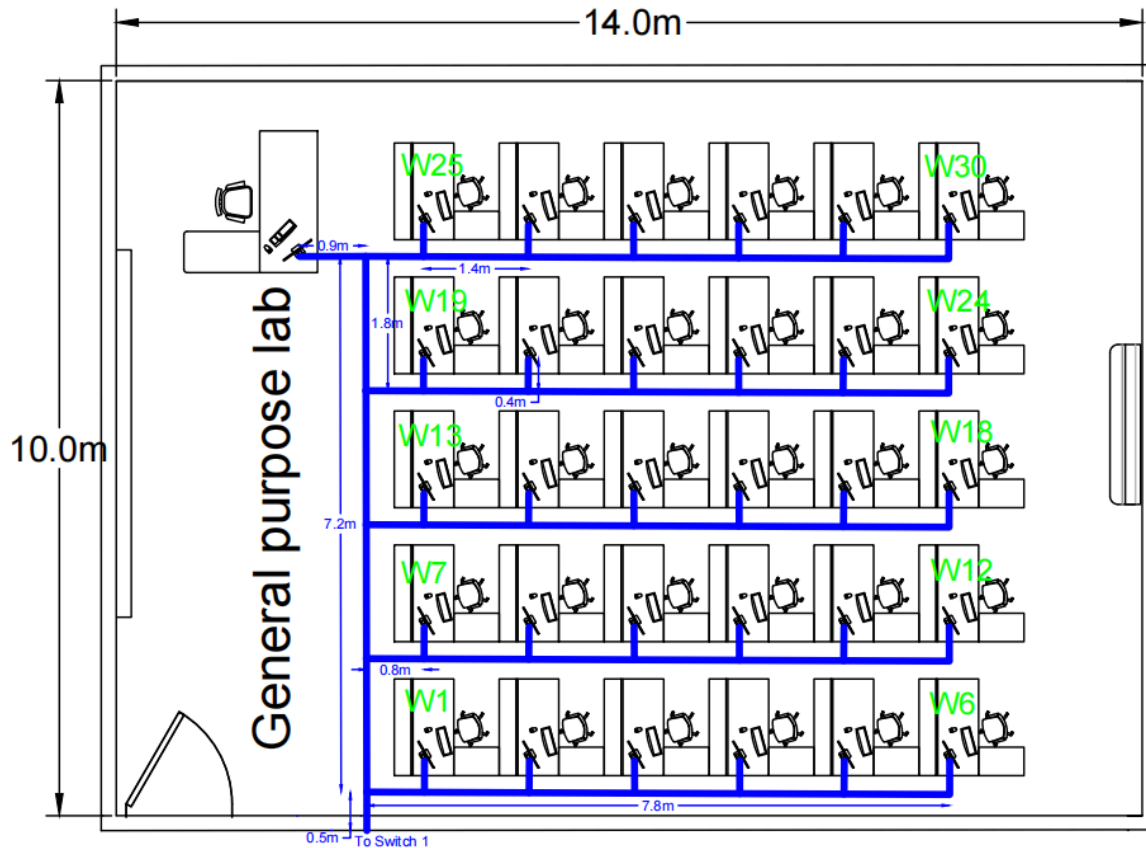


Figure 2.2.2.1: Room Level Cabling - General Purpose Lab

Connection to Switch 1 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Workstation W1	Cat 6	$0.6 + 9.1 + 5.8 + 2.8 + 0.5 + 0.8 + 0.4 = 20.0$	0.75	20.75
Workstation W2	Cat 6	$20.0 + 1.4 = 21.4$	0.75	22.15
Workstation W3	Cat 6	$21.4 + 1.4 = 22.8$	0.75	23.55
Workstation W4	Cat 6	$22.8 + 1.4 = 24.2$	0.75	24.95
Workstation W5	Cat 6	$24.2 + 1.4 = 25.6$	0.75	26.35
Workstation W6	Cat 6	$25.6 + 1.4 = 27.0$	0.75	27.75
Workstation W7	Cat 6	$20.0 + 1.8 = 21.8$	0.75	22.55
Workstation W8	Cat 6	$21.8 + 1.4 = 23.2$	0.75	23.95

Workstation W9	Cat 6	$23.2 + 1.4 = 24.6$	0.75	25.35
Workstation W10	Cat 6	$24.6 + 1.4 = 26.0$	0.75	26.75
Workstation W11	Cat 6	$26.0 + 1.4 = 27.4$	0.75	28.15
Workstation W12	Cat 6	$27.4 + 1.4 = 28.8$	0.75	29.55
Workstation W13	Cat 6	$21.8 + 1.8 = 23.6$	0.75	24.35
Workstation W14	Cat 6	$23.6 + 1.4 = 25.0$	0.75	25.75
Workstation W15	Cat 6	$25.0 + 1.4 = 26.4$	0.75	27.15
Workstation W16	Cat 6	$26.4 + 1.4 = 27.8$	0.75	28.55
Workstation W17	Cat 6	$27.8 + 1.4 = 29.2$	0.75	29.95
Workstation W18	Cat 6	$29.2 + 1.4 = 30.6$	0.75	31.35
Workstation W19	Cat 6	$23.6 + 1.8 = 25.4$	0.75	26.15
Workstation W20	Cat 6	$25.4 + 1.4 = 26.8$	0.75	27.55
Workstation W21	Cat 6	$26.8 + 1.4 = 28.2$	0.75	28.95
Workstation W22	Cat 6	$28.2 + 1.4 = 29.6$	0.75	30.35
Workstation W23	Cat 6	$29.6 + 1.4 = 31.0$	0.75	31.75
Workstation W24	Cat 6	$31.0 + 1.4 = 32.4$	0.75	33.15
Workstation W25	Cat 6	$25.4 + 1.8 = 27.2$	0.75	27.95
Workstation W26	Cat 6	$27.2 + 1.4 = 28.6$	0.75	29.35
Workstation W27	Cat 6	$28.6 + 1.4 = 30.0$	0.75	30.75
Workstation W28	Cat 6	$30.0 + 1.4 = 31.4$	0.75	32.15
Workstation W29	Cat 6	$31.4 + 1.4 = 32.7$	0.75	33.45
Workstation W30	Cat 6	$32.7 + 1.4 = 34.2$	0.75	34.95
Teaching Desk	Cat 6	$0.6 + 9.1 + 5.8 + 2.8 + 0.5 + 7.2 + 0.9 = 26.9$	0.75	27.65
WAP	Cat 6	$0.6 + 9.1 + 5.8 + 8.7 + 3.7 = 27.9$	2.7	30.6
Total		867.7	25.95	893.65

2.2.2.2 Embedded Lab

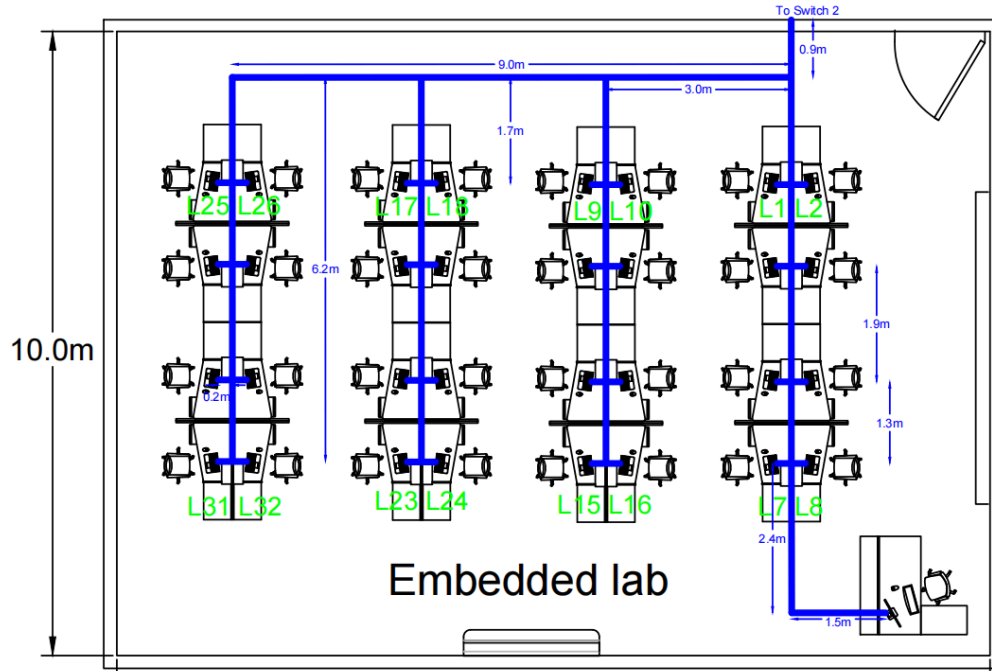


Figure 2.2.2.2: Room Level Cabling - Embedded Lab

Connection to Switch 2 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Lab Desk L1	Cat 6	$0.6 + 1.4 + 6.3 + 4.9 + 2.8 + 0.9 + 1.7 + 0.2 = 18.8$	0.75	19.55
Lab Desk L2	Cat 6	18.8	0.75	19.55
Lab Desk L3	Cat 6	$18.8 + 1.3 = 20.1$	0.75	20.85
Lab Desk L4	Cat 6	20.1	0.75	20.85
Lab Desk L5	Cat 6	$20.1 + 1.9 = 22.0$	0.75	22.75
Lab Desk L6	Cat 6	22.0	0.75	22.75
Lab Desk L7	Cat 6	$22.0 + 1.3 = 23.3$	0.75	24.05
Lab Desk L8	Cat 6	23.3	0.75	24.05
Lab Desk L9	Cat 6	$18.8 + 3.0 = 21.8$	0.75	22.55
Lab Desk L10	Cat 6	21.8	0.75	22.55

Lab Desk L11	Cat 6	$21.8 + 1.3 = 23.1$	0.75	23.85
Lab Desk L12	Cat 6	23.1	0.75	23.85
Lab Desk L13	Cat 6	$23.1 + 1.9 = 25.0$	0.75	25.75
Lab Desk L14	Cat 6	25.0	0.75	25.75
Lab Desk L15	Cat 6	$25.0 + 1.3 = 26.3$	0.75	27.05
Lab Desk L16	Cat 6	26.3	0.75	27.05
Lab Desk L17	Cat 6	$21.8 + 3.0 = 24.8$	0.75	25.55
Lab Desk L18	Cat 6	24.8	0.75	25.55
Lab Desk L19	Cat 6	$24.8 + 1.3 = 26.1$	0.75	26.85
Lab Desk L20	Cat 6	26.1	0.75	26.85
Lab Desk L21	Cat 6	$26.1 + 1.9 = 28.0$	0.75	28.75
Lab Desk L22	Cat 6	28.0	0.75	28.75
Lab Desk L23	Cat 6	$28.0 + 1.3 = 29.3$	0.75	30.05
Lab Desk L24	Cat 6	29.3	0.75	30.05
Lab Desk L25	Cat 6	$24.8 + 3.0 = 27.8$	0.75	28.55
Lab Desk L26	Cat 6	27.8	0.75	28.55
Lab Desk L27	Cat 6	$27.8 + 1.3 = 29.1$	0.75	29.85
Lab Desk L28	Cat 6	29.1	0.75	29.85
Lab Desk L29	Cat 6	$29.1 + 1.9 = 31.0$	0.75	31.75
Lab Desk L30	Cat 6	31.0	0.75	31.75
Lab Desk L31	Cat 6	$31.0 + 1.3 = 32.3$	0.75	33.05
Lab Desk L32	Cat 6	32.3	0.75	33.05
Teaching Desk	Cat 6	$0.6 + 1.4 + 6.3 + 4.9 + 2.8 + 0.9 + 1.7 + 1.3 + 1.9 + 1.3 + 2.4 + 1.5 = 27.0$	0.75	27.75
WAP	Cat 6	$0.6 + 1.4 + 6.3 + 4.9 + 3.6 = 24.5$	2.7	27.2
Total		869.1	27.45	896.55

2.2.2.3 Video Conferencing Room

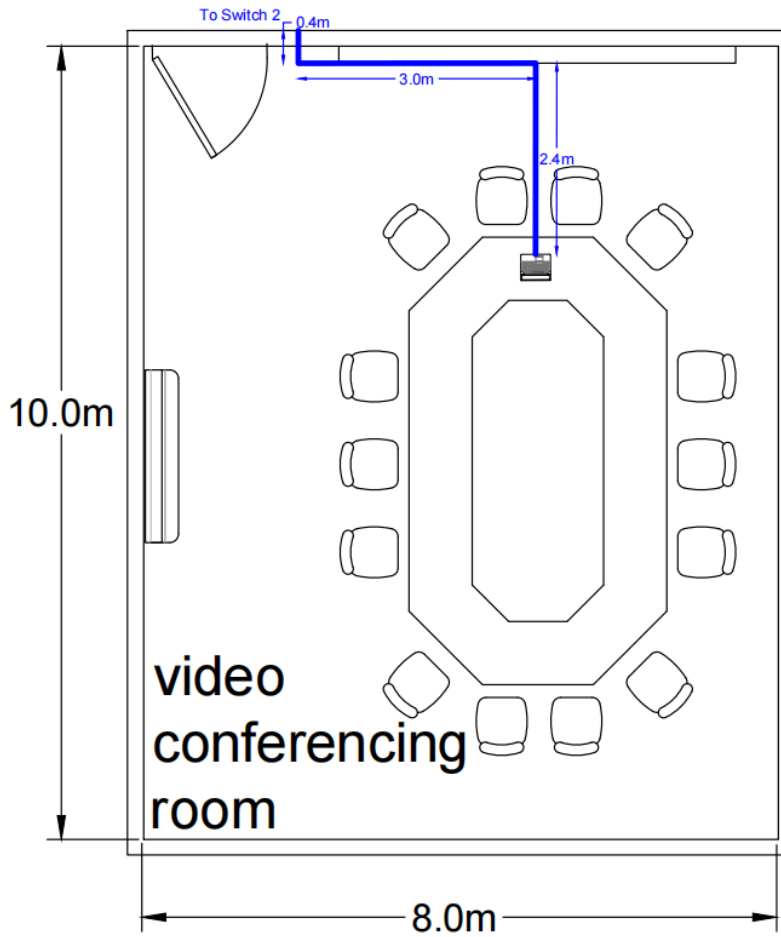


Figure 2.2.2.3: Room Level Cabling - Video Conferencing Room

Connection to Switch 2 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Laptop	Cat 6	$0.6 + 1.4 + 6.3 + 4.9 + 5.2 + 2.8 + 0.4 + 3.0 + 2.4 = 27.0$	0.75	27.75
WAP	Cat 6	$0.6 + 1.4 + 6.3 + 4.9 + 5.2 + 7.1 + 2.4 = 27.9$	2.7	30.6
Total		54.9	3.45	58.35

2.2.2.4 Hybrid Room

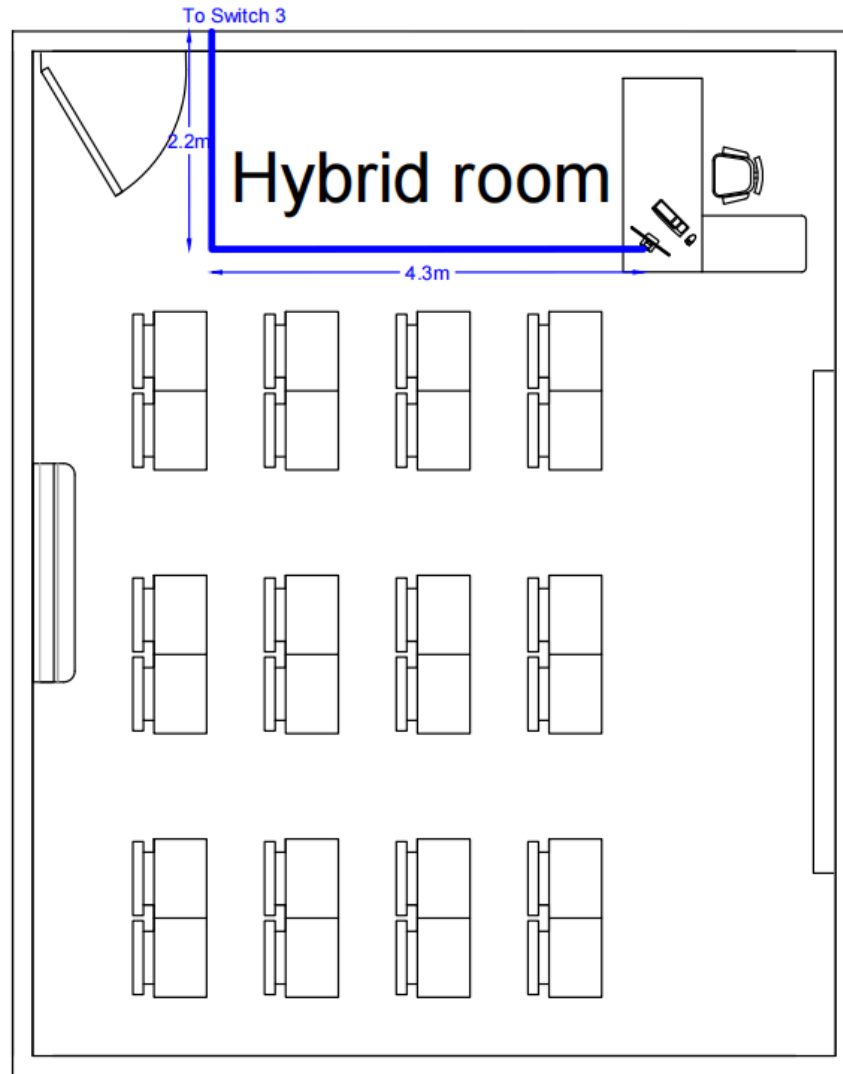


Figure 2.2.2.4: Room Level Cabling - Hybrid Room

Connection to Switch 3 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Teaching Desk	Cat 6	$0.6 + 6.3 + 4.9 + 5.2 + 8.1 + 2.8 + 2.2 + 4.3 = 34.4$	0.75	35.15
WAP	Cat 6	$0.6 + 6.3 + 4.9 + 5.2 + 8.1 + 7.7 + 0.8 = 33.6$	2.7	36.3
Total		68.0	3.45	71.45

2.2.2.5 Printing Area & Small Meeting Room

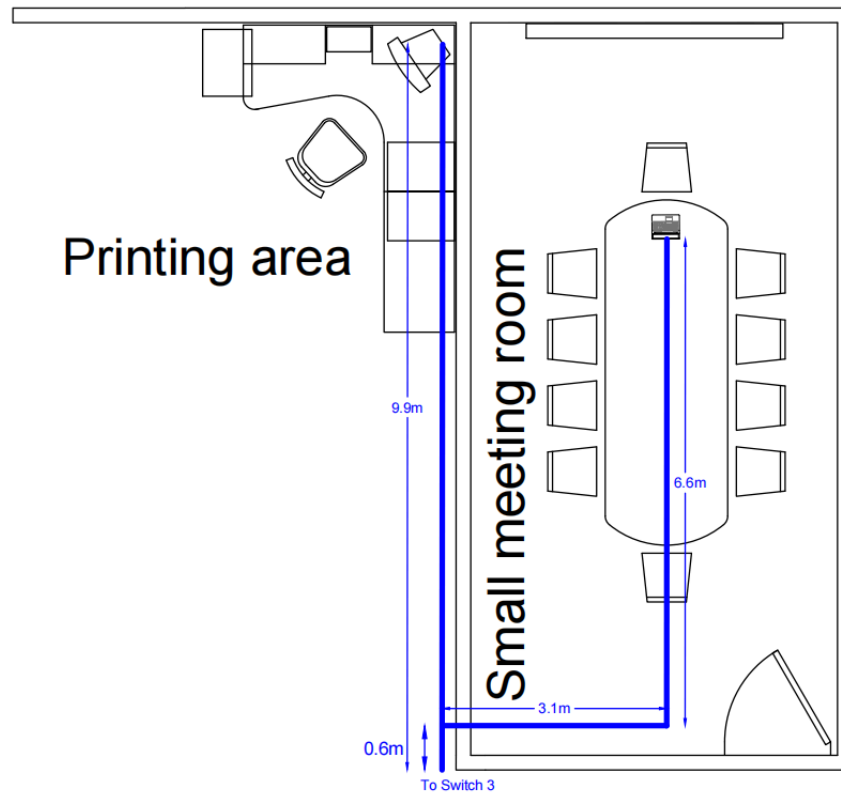


Figure 2.2.2.5: Room Level Cabling - Printing Area & Small Meeting Room

Connection to Switch 3 with Patch Panel	Cable	Horizontal Length (m)	Vertical Length (m)	Total Length (m)
Printing	Cat 6	$0.6 + 6.3 + 4.9 + 5.2 + 8.1 + 7.1 + 2.8 + 0.6 + 9.9 = 45.5$	0.75	46.25
Laptop	Cat 6	$0.6 + 6.3 + 4.9 + 5.2 + 8.1 + 7.1 + 2.8 + 0.6 + 3.1 + 6.6 = 45.3$	0.75	46.05
WAP	Cat 6	$0.6 + 6.3 + 4.9 + 5.2 + 8.1 + 8.4 + 7.2 = 40.7$	2.7	43.4
Total		131.5	4.2	135.7

3.0 Number of connections, patch cords and switch ports needed

For this network design, each end device is assumed to require one wired Ethernet connection. All horizontal cabling uses CAT6 UTP, terminating at wall data outlets in the work areas and patch panels in the Intermediate Distribution Facility (IDF). Each connection requires two patch cords: one connecting the device to the wall outlet and one connecting the patch panel to the network switch in the IDF. All network switches are centralized in the IDF on each floor, while wireless access points are installed within each room. Fiber optic cabling is assumed for backbone connections between floors and is not included in the horizontal cabling calculations.

3.1 Ground Floor

Area	Number of connections needed
Cisco Networking Lab	32
General Purpose Lab	32
Lecturer Office	7

Table 3.1.1 Number of connections needed

Area	Number of patch cords needed
Cisco Networking Lab	64
General Purpose Lab	64
Lecturer Office	14

Table 3.1.2 Number of patch cords needed

Area	Number of switch ports needed
Cisco Networking Lab	32
General Purpose Lab	32
Lecturer Office	7

Table 3.1.3 Number of switch ports needed

3.2 First Floor

Area	Number of connections needed
General Purpose Lab	32
Embedded Lab	34
Video Conferencing Room	2
Hybrid Room	2
Printing Area & Small Meeting Room	3

Table 3.2.1 Number of connections needed

Area	Number of patch cords needed
General Purpose Lab	64
Embedded Lab	68
Video Conferencing Room	4
Hybrid Room	4
Printing Area & Small Meeting Room	6

Table 3.2.2 Number of patch cords needed

Area	Number of switch cords needed
General Purpose Lab	32
Embedded Lab	34
Video Conferencing Room	2
Hybrid Room	2
Printing Area & Small Meeting Room	3

Table 3.2.3 Number of switch cords needed

3.3 Summary of both floor

Area	Number of connections needed
Cisco Networking Lab	32
General Purpose Lab	32
Lecturer Office	7
General Purpose Lab	32
Embedded Lab	34
Video Conferencing Room	2
Hybrid Room	2
Printing Area & Small Meeting Room	3
Total	144

Table 3.3.1 Number of connections needed

Area	Number of patch cords needed
Cisco Networking Lab	64
General Purpose Lab	64
Lecturer Office	14
General Purpose Lab	64
Embedded Lab	68
Video Conferencing Room	4
Hybrid Room	4
Printing Area & Small Meeting Room	6
Total	288

Table 3.3.2 Number of patch cords needed

Area	Number of switch cords needed
Cisco Networking Lab	32
General Purpose Lab	32
Lecturer Office	7
General Purpose Lab	32
Embedded Lab	34
Video Conferencing Room	2
Hybrid Room	2
Printing Area & Small Meeting Room	3
Total	144

Table 3.3.3 Number of switch cords needed

4.0 Identify cables types and length

The design of the building network utilizes a strategic, two-pronged cabling approach to ensure both high-speed performance within the labs and reliable long-distance connectivity to the campus core. These cable types are clearly labeled in the network distribution diagrams to distinguish between distribution and backbone connections.

In this network design, 12-core Category 6 (Cat 6) UTP is used for all wired Distribution Cabling connections within the work areas. This includes connections from workstations, wireless access points, printers, and other end devices to wall data outlets, as well as connections from patch panels to network switches located in the Intermediate Distribution Facility (IDF). Cat 6 UTP provides excellent performance for stable 1 Gigabit Ethernet (1000BASE-T) transmission and complies with the 100-meter Ethernet distance limitation, making it suitable for high-density environments such as laboratories and classrooms. The designation of 12-core cable is a strategic choice, providing high cable density and built-in spare capacity or allowing for dual permanent links to critical workstations, thus enhancing the overall cabling redundancy. While it supports 10 Gigabit Ethernet (10GBASE-T), its capability at this speed is limited to runs under 55 meters. This choice offers a significant cost-saving advantage, allowing us to allocate more budget toward high-performance active devices (switches/router) while still providing sufficient speed and reliability at the desktop level. For budget planning, the cost is estimated at RM 4.00 per meter.

For inter-floor and building-level connectivity, a specialized OM3 Multimode Fiber Optic Cable is used as the Backbone Cabling to connect the IDFs on each floor to the Main Distribution Facility (MDF). Fiber optic cabling provides crucial benefits: higher bandwidth, longer transmission distance, and complete resistance to electromagnetic interference, ensuring reliable communication between floors. By choosing OM3, we ensure the backbone can readily support 10 Gigabit Ethernet (10GBASE-SR) over runs up to 300 meters, allowing for smooth handling of all aggregated traffic and providing a clear upgrade path to 40 Gbps or 100 Gbps for future network expansion. The strategic use of this media guarantees that the core infrastructure meets the highest performance and scalability goals of the project. For budget planning, the cost is estimated at RM 6.00 per meter.

4.1 CAT6 twisted cable

Area	Length (m)		Total Length (m)	Total Price of Cable Used (RM)
	Horizontal	Vertical		
Ground Floor				
Cisco Networking Lab	455	25.2	480.2	1920.8
General Purpose Lab	387.1	25.95	413.05	1652.2
Lecturer Office	86.9	7.2	94.1	376.4
Ground Floor Room Total			987.35	3949.4
First Floor				
General Purpose Lab	867.7	25.95	893.65	3574.6
Embedded Lab	869.1	27.45	896.55	3586.2
Video Conferencing Room	54.9	3.45	58.35	233.4
Hybrid Room	68.0	3.45	71.45	285.8
Printing Area & Small Meeting Room	131.5	4.2	135.7	542.8
First Floor Room Total			2055.7	8222.8

Table 4.1 Total length and prices of CAT6 twisted cable used

4.2 Fiber Optic Cables

Path	Length (m)		Total Length (m)	Price of cables used (RM)
	Horizontal	Vertical		
Ground Floor				
Switch 1 to server	2.4	0.5	2.9	17.4
Switch 2 to server	1.7	0.5	2.2	13.2
Switch 3 to server	1.0	0.5	1.5	9
Total			6.6	39.6
First Floor				
Switch 1 to server	34.9	1.7	36.6	219.6
Switch 2 to server	33.5	1.7	35.2	211.2
Switch 3 to server	32.1	1.7	33.8	202.8
Total			105.6	633.6

Table 4.2 Total length and prices of fiber optic cable used

4.3 Summary of Cables Used

Types of cables	Length (m)	Price of cables used (RM)
Ground Floor		
CAT 6 Twisted Cable	987.35	3949.4
Fiber Optic Cable	6.6	39.6
Total	993.95	3989
First Floor		
CAT 6 Twisted Cable	2055.7	8222.8
Fiber Optic Cable	105.6	633.6
Total	2161.3	8856.4

Ground Floor & First Floor Cables Used	
Length (m)	Price (RM)
3155.25	12845.4

Table 4.3 Total length and prices of cable used

5.0 Meeting Minutes

5.1 Meeting Minutes#1

DATE/TIME		11 December 2025 19:00	
LOCATION		Online(Google Meet)	
AGENDA		1. Analysis Task 4 requirements 2. Identify work area arrangement and IDF device placement. 3. Design the Network Distribution Diagram 4. Research suitable type of cables 5. Task distribution	
Meeting MC		Tey Xin Yi	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Tey Xin Yi		19:00	
Lee Hui Zhen		19:00	
Tan Hui Shan		19:00	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Understanding Task 4 Requirements	- Tey explained that Task 4 need to sketch pc and network distribution diagram and identify: - connections, patch cord, switch cord - Cable length and types	Tey (11/12)
2	Identifying work area arrangement and IDF device placement	- All members discussed the arrangement of work areas based on the approved floor plan. - Lee proposed placing all switches and patch panels in the Electrical Room and designating it as the IDF. - Tey highlighted that centralized IDF placement would simplify maintenance and troubleshooting.	Lee & Tey (11/12)

3	Design the Network Distribution Diagram	• All members discussed the structure of the network distribution diagram at both room and building levels. • Tey suggested clearly separating horizontal cabling and backbone connections. • Tan emphasized the importance of labeling cable types and connection paths clearly.	Tey & Tan (11/12)
4	Research suitable type of cables	• All members discussed suitable cable types for Task 4. • Tan suggested using CAT6 UTP for horizontal cabling due to its performance and cost efficiency. • Lee proposed using fiber optic cables for backbone connections between floors.	Tan & Lee (11/12)
5	Task Distribution	• Task distribution assigned as below: 1. Tey a. Identify work area of Ground Floor b. Designing network distribution diagram (Building & Room Level) of Ground Floor 2. Lee a. Identify work area of First Floor b. Designing network distribution diagram (Building & Room Level) of First Floor 3. Tan a. Identify number of connections, patch cords, switch cords b. Identify cables types used, length and price c. Prepare meeting minutes	All members (11/12)
6	Meeting ended	21:00	

5.2 Meeting Minutes#2

DATE/TIME		14 December 2025 11:00	
LOCATION		Online(Google Meet)	
AGENDA		1. Review current progress and floor-specific designs. 2. Finalize Network Distribution Diagram (Building & Room Level) 3. Check calculation of cable lengths and price	
Meeting MC		Tan Hui Shan	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Tey Xin Yi		11:00	
Lee Hui Zhen		11:00	
Tan Hui Shan		11:00	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Review current progress and floor-specific designs	- All members reviewed the current progress of Task 4. - Tey pointed out minor inconsistencies in floor-specific layouts that required adjustment. - Lee suggested standardizing labels for all work areas and network devices.	Tey (14/12)
2	Finalize Network Distribution Diagram	- All members discussed and finalized the room-level and building-level network distribution diagrams. - Tan confirmed that all CAT6 and fiber connections were correctly represented. - Lee ensured the diagrams matched the scaled floor plans.	Lee & Tan (14/12)

3	Check calculation of cable lengths and price	• All members reviewed the cable length calculations based on the floor plan scale. • Tey suggested adding extra length for vertical runs and corners. • Tan verified estimated cable costs to ensure the design remained within budget.	Tey & Tan (14/12)
4	Meeting ended	14:00	